Practice Set 6 Paper 3 – Subspecialities of Urology

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Practice paper compiled from the question bank.

Q1. Discuss the management of B/l VUR and discuss its long term outcome

Type: LAQ (25 marks) (also answered in Practice Set 2 Paper 3, Q2, 2015-Summer Paper 3, Q2 and 2020-Summer Paper 1, Q2; long-term outcome added here)

Introduction

- **VUR (vesicoureteral reflux):** retrograde flow of urine from the bladder to the upper tract. **Primary:** short intravesical tunnel; tunnel length to ureteric diameter should be **5:1** (Paquin). **Secondary:** high bladder pressure (PUV (posterior urethral valves), neurogenic bladder, BBD (bladder and bowel dysfunction)).
- **Bilateral VUR is a high-risk feature:** it resolves less and scars more. **Infection, not sterile reflux, damages the kidney**, so management targets infection risk, not the reflux itself.

International grading (VCUG, voiding cystourethrogram)

Grade	Findings
I	Ureter only, not dilated
II	Pelvis and calyces reached, no dilatation
III	Mild–moderate dilatation, minimal forniceal blunting
IV	Moderate dilatation and tortuosity
V	Gross dilatation; papillary impressions lost

Evaluation

- Febrile UTI (urinary tract infection) history, BBD, BP (blood pressure), creatinine, proteinuria.
- **RBUS (renal and bladder ultrasound)**; **VCUG** (grade; urethra to exclude PUV); **DMSA (dimercaptosuccinic acid) scan** for scars and split function; nuclear cystogram for follow-up. **UDR (ureteral diameter ratio)** predicts resolution.

Natural history and risk

- Resolution: grade II 63–85%, grade III about 50%; grades IV–V and bilateral III rarely. **Bilateral grade IV persists most.**
- **High risk (Hidas):** female, under 1 year, **bilateral**, grade III–V, BBD, scars, multiple febrile UTIs. Two-year breakthrough UTI on CAP (continuous antibiotic prophylaxis): 8.6% low, 26% intermediate, 62.5% high risk.

Medical management

- **Treat BBD first:** timed voiding, hydration, constipation treatment, anticholinergics if needed.
- **CAP, one low dose at night:** trimethoprim or amoxicillin under 2 months (avoid sulfonamide: neonatal jaundice); TMP-SMX (trimethoprim-sulfamethoxazole) or nitrofurantoin after 2 months.
- **RIVUR (2014):** recurrent UTI **23.6% → 12.9%**; **scarring unchanged**. **Swedish Reflux Trial** (grade III–IV, 1–2 years): febrile UTI in girls 19% on CAP, 23% after endoscopy, 57% on surveillance; boys no difference. **IRSC (International Reflux Study in Children) and Birmingham:** surgery equalled medical care for new scars.
- **Also know:** circumcision; EAU/ESPU (European Association of Urology/European Society for Paediatric Urology): CAP for high risk.

Indications for correction

- Breakthrough febrile UTI despite good bladder and bowel habits; new scars or falling function; persistent bilateral grade IV–V; poor adherence; parental choice. Asymptomatic bilateral high-grade VUR in an older boy with normal kidneys may be observed (controversial).

Surgical options and selection

Option	Key point	Best for
Endoscopic Dx/HA (dextranomer/hyaluronic acid)	HIT (hydrodistension implantation) inside the orifice beats STING (subureteric injection); double HIT about 90%; recurrence 20% at 2 years	Grades II–IV, least invasive
Cohen cross-trigonal	Success over 98%; later ureteroscopy hard	**Bilateral VUR:** both sides through one cystotomy; trabeculated bladder
Politano–Leadbetter	New superomedial hiatus, long tunnel	Grade V, ± tapering
Lich–Gregoir (extravesical)	No haematuria or spasm; **retention up to 10% after bilateral repair** (pelvic plexus injury)	Unilateral
RALUR-EV (robot-assisted extravesical)	Less pain	Unilateral, older child

- Open reimplantation success **95.9%** overall. Secondary VUR: treat the bladder first.

Long-term outcome

Outcome	Facts
Renal scarring	About 15% get a febrile UTI within 2 years; about 15% of those scar. Risk highest under 1 year. Much "scarring" in boys with high grade is **congenital dysplasia**, not preventable by any treatment
Hypertension	Reflux nephropathy may be the commonest cause of childhood hypertension after essential hypertension (renin–angiotensin). **34–38%** of scarred children are hypertensive as adults. Correcting VUR does not lower BP; nephrectomy of a small scarred kidney helps selected cases
CKD (chronic kidney disease) and ESKD (end-stage kidney disease)	Now rare: RRT (renal replacement therapy) from reflux nephropathy fell from 1.9 to 0.6 per million (USRDS (United States Renal Data System), 1996–2014). Bilateral scars cause hyperfiltration, proteinuria, concentrating defect, tubular acidosis
Pregnancy	VUR in late pregnancy: pyelonephritis about one-third versus under 5%. Scars: about **3× hypertension, 8× pre-eclampsia**; bilateral worse. Impaired function: fetal loss. Reimplanted women still get UTIs
Growth; offspring	Catch-up growth after infection control; about 43% of screened offspring have VUR

- **Concept:** in adult life **the scar, not the reflux, carries the risk**, so correction does not remove it. Scarred patients need lifelong BP, proteinuria and creatinine checks.

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Treat BBD -> low risk: observe | high risk (bilateral): CAP
Breakthrough UTI / new scar / persistent IV-V -> Dx/HA or Cohen
Scars -> lifelong BP and proteinuria checks
```

Diagram to draw: grades I–V; Cohen cross-trigonal reimplant.

Key points to remember:

- Bilateral grade IV persists most; RIVUR: CAP halves UTI, not scarring.
- Cohen (over 98%) suits bilateral VUR; bilateral extravesical repair risks retention up to 10%.
- 34–38% of scarred children become hypertensive; correction does not lower BP.
- Scars bring 8× pre-eclampsia risk: the scar, not the reflux, matters.

Q2. Discuss development of urethra. Classify congenital anomalies and discuss management of epispadias

Type: LAQ (25 marks) (isolated penopubic epispadias also answered in 2022-Summer Paper 3, Q3f)

Development of the urethra

1. Division of the cloaca (weeks 5–6)

- The urorectal septum divides the cloaca into the dorsal anorectal canal and the ventral **UGS (urogenital sinus)**.
- Wolffian duct entry is the landmark. Above it, the **vesicourethral canal** forms the bladder and pelvic urethra. Below it, the **definitive UGS** forms the phallic urethra in males, and the urethra and vestibule in females.
- UGS also forms the prostate (buds at 10 weeks) and Cowper and Littre glands; the **utricle** is Müllerian.
- The epithelium of the **entire male urethra is endodermal** (UGS origin).

2. Penile urethra: the "double zipper" (androgen-dependent)

- Indifferent genital tubercle until 8–9 weeks. Testosterone (Leydig cells, 8 weeks) is converted to DHT (dihydrotestosterone) by **type 2 5α-reductase**.
- The **urethral plate** (endoderm) forms in the tubercle.
- **Opening zipper:** the plate canalises proximal to distal into a urethral groove bounded by the urethral folds.
- **Closing zipper:** the folds fuse in the ventral midline, forming the tubular urethra and median raphe.
- **Glanular urethra:** formed by direct canalisation of the plate inside the glans (current endodermal view; Glenister's ectodermal ingrowth theory is outdated).
- **Timetable:** coronal sulcus 11–12 weeks; fusion complete 16 weeks; urethra complete 18 weeks; fetal curvature gone 20 weeks.
- The prepuce forms with the glanular urethra, so hypospadias has a hooded foreskin.
- **Female:** the vestibular plate canalises but the folds do not fuse, giving an open vestibule and labia minora.

Classification of congenital urethral anomalies

Mechanism	Anomaly
Failed ventral fold fusion	**Hypospadias** (1 in 250 males; commonest at shaft–glans junction); congenital urethral fistula
Dorsal defect	**Epispadias** (exstrophy–epispadias complex)
Obstruction	**PUV (posterior urethral valves)**, Young type 1 (95%) and type 3; **anterior urethral valves** with diverticulum; urethral atresia or hypoplasia; meatal stenosis
Duplication	**Urethral duplication**: sagittal; ventral channel is functional; Effmann type I blind, II complete, III with caudal duplication
Dilatation	**Megalourethra**: scaphoid (spongiosum deficient) or fusiform (all erectile tissue deficient, more severe); prune-belly association
Cystic or remnant	Cowper's syringocele; enlarged utricle (proximal hypospadias); parameatal cyst

Epispadias: definition and classification

- A **dorsal urethral defect**: the roof is replaced by a broad mucosal strip. Cause: absent mesodermal reinforcement of the cloacal membrane; the genital tubercles fuse caudal to it.

- **Male (1 in 117,000):** glanular, penile, **penopubic** (about 70%, incontinent). **Female (1 in 484,000), Davis:** vesicular (patulous), subsymphyseal, retrosymphyseal (incontinent).
- **Features:** short broad penis with **dorsal chordee**; pubic diastasis (less than exstrophy); small bladder from no outlet resistance; **VUR (vesicoureteral reflux) 30–40% (male), 30–75% (female)**. Female: bifid clitoris, depressed mons.

Evaluation

- Ultrasound; VCUG (voiding cystourethrogram); cystoscopy and capacity under anaesthesia.
- **Key decision:** epispadias or exstrophy pathway. Treat as exstrophy (osteotomy and closure) if pubic diastasis is over 3 cm, the intrasymphyseal bar is poor, sphincter is absent or capacity is tiny.

Management of epispadias

- **Aims:** continence, safe upper tracts, a straight functional penis of good length.
- **Testosterone** 2 mg/kg intramuscularly, 5 and 2 weeks before urethroplasty.

Stage 1: penile reconstruction and urethroplasty (6–12 months)

- Release chordee: divide suspensory ligaments and scar; free corpora from the inferior pubic rami.
- **Modified Cantwell–Ransley:** outline an 18-mm plate to the tip; reverse MAGPI (meatal advancement and glanuloplasty) brings the meatus to the tip; free the plate from the corpora (distal 1 cm left on the glans); tubularise over 8 Fr; transpose it **ventral to the corpora**, which are rotated medially ± cavernocavernostomy; close glans wings; Z-plasty at the base; stent 10–12 days.
- **Mitchell–Bägli complete disassembly:** reliable chordee correction, but often leaves hypospadias.

Stage 2: bladder neck reconstruction (4–5 years, capacity about 100 mL)

- **YDL (Young–Dees–Leadbetter)** repair + Marshall–Marchetti–Krantz suspension + ureteric reimplantation. Osteotomy usually unnecessary.
- **Good capacity:** single-stage repair possible.

Female epispadias: at 6–12 months excise the urethral roof, close over 10 Fr, unite clitoral halves and labia minora, monsplasty; YDL at 4–5 years.

Outcomes and complications

- Continence after bladder neck reconstruction: **males 47–82%, females 70–87%**; capacity is the main predictor.
- Early fistula up to one-fifth (at the penile base); long-term fistula and stricture about 5%; erections and paternity preserved.
- Better than exstrophy (bladder never exposed). Failures: bulking injection, augmentation, continent stoma.

Diagram to draw: double zipper (canalisation then fusion); penopubic epispadias.

> **Key points to remember:**
> - Whole male urethra is endodermal UGS; fold fusion complete at 16, urethra at 18 weeks.
> - About 70% of male epispadias are penopubic and incontinent; diastasis over 3 cm → exstrophy pathway.
> - Cantwell–Ransley at 6–12 months; YDL at about 100 mL capacity.
> - Continence after YDL: males 47–82%, females 70–87%.

Q3a. Various lasers in treatment of BPH

Type: SAQ (10 marks) (also answered in 2017-Winter Paper 2, Q3a)

Principle

- LASER (light amplification by stimulated emission of radiation) turns light into local heat. The **chromophore** (water or haemoglobin) and wavelength decide **penetration depth**.
- About 60 °C coagulates (delayed sloughing); 100 °C or more vaporises. Coagulation lasers are obsolete.

Lasers used in BPH (benign prostatic hyperplasia)

Laser	Wavelength	Chromophore	Penetration	Procedure
Ho:YAG (holmium:yttrium-aluminium-garnet), pulsed	2120 nm	Water	0.4 mm	**HoLEP** (holmium laser enucleation of the prostate)
Thulium:YAG (continuous) / **TFL** (thulium fibre laser)	2013 / 1940 nm	Water (TFL at absorption peak, about 4× holmium)	0.2 mm	ThuLEP (enucleation), ThuVEP (vapoenucleation), ThuVARP (vaporesection)
KTP (potassium titanyl phosphate) / LBO (lithium triborate)	532 nm	Haemoglobin	0.8 mm	**PVP** (photoselective vaporisation), 120–180 W
Diode	Various	Mixed	0.5–5 mm	Vaporisation
Nd:YAG (neodymium)	1064 nm	—	About 1 cm	Abandoned (sloughing)

- **KTP:** Nd:YAG beam halved to 532 nm by a crystal; haemoglobin absorbs it, so vaporisation is bloodless.

Evidence

- **HoLEP:** endoscopic equivalent of open prostatectomy; **size-independent**. The only endoscopic procedure **superior to TURP** (transurethral resection of the prostate) for IPSS (International Prostate Symptom Score) and Qmax. Less bleeding, shorter catheter time; steep learning curve.
- **ThuLEP:** equals HoLEP.
- **GOLIATH** (180 W PVP vs TURP, about 270 men, ≤100 cc): PVP non-inferior at 2 years; shorter catheter time and stay. Retreatment high above 80 cc.
- **AUA (American Urological Association):** HoLEP or ThuLEP for any size; PVP for under 80 cc; HoLEP, ThuLEP or PVP for **higher bleeding risk** (anticoagulated men).

Complications

- Dysuria and storage symptoms (worst after PVP); transient stress incontinence; retrograde ejaculation in most; bladder neck contracture 0–3.2%; stricture; morcellator bladder injury. No TUR (transurethral resection) syndrome (saline).
- **Safety:** wavelength-specific eyewear for all; laser safety officer.

Diagram to draw: HoLEP: adenoma enucleated to the capsule, pushed into bladder, morcellated.

Key points to remember:

- Penetration: thulium 0.2, holmium 0.4, KTP 0.8 mm, Nd:YAG 1 cm.
- HoLEP is size-independent and superior to TURP.
- GOLIATH: PVP non-inferior to TURP; avoid PVP above 80 cc.

Q3b. Telomerase

Type: SAQ (10 marks)

Telomeres

- **Telomeres** are repetitive sequences (TTAGGG repeats) at chromosome ends. They protect ends from degradation and fusion.
- DNA polymerase cannot copy the very end of the lagging strand (end-replication problem). So about **100 base pairs are lost per division**: the "mitotic clock".
- Critical shortening triggers **replicative senescence** (Hayflick limit).

Telomerase: structure and function

- A specialised **reverse transcriptase** that adds telomeric repeats.
- Two core parts: **hTERT** (human telomerase reverse transcriptase, catalytic protein) and **hTR** (telomerase RNA, the template).
- **Normal expression:** only rapidly dividing tissues: germ cells (testis), intestinal epithelium, endometrium, activated lymphocytes. Most somatic cells lack it.

Role in cancer

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Proliferation -> moderate shortening -> senescence/apoptosis (barrier 1)
-> checkpoint lost (p53) -> severe shortening -> chromosomal instability (barrier 2)
-> telomerase activation -> immortal cancer cell
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- **85–90% of human tumours** express high telomerase, giving unlimited replication.
- **ALT (alternative lengthening of telomeres):** telomerase-independent recombination; linked to ATRX or DAXX loss.
- **TERT promoter mutations** switch on hTERT.

Urological relevance

Organ	Finding
Bladder	TERT promoter mutation is the commonest mutation (over 60% even in PUNLMP, papillary urothelial neoplasm of low malignant potential). Urine markers: telomerase activity by **TRAP (telomeric repeat amplification protocol)**, a PCR (polymerase chain reaction)-based assay; hTERT messenger RNA; AssureMDx (FGFR3, TERT, HRAS mutations plus methylation). Inflammation gives false positives; not routine
Upper tract	hTR positive in almost all UTUC (upper tract urothelial carcinoma); rises with stage
Prostate	Short telomeres in cancer and premalignant lesions; short tumour telomeres predict recurrence; variable stromal telomeres predict cancer death

- **Also know:** senescent cells secrete SASP (senescence-associated secretory phenotype) cytokines in ageing and BPH (benign prostatic hyperplasia).

Therapy

- **Imetelstat** (telomerase inhibitor) is used in myelodysplastic syndrome.
- hTERT-based vaccines (UV1, GX301, INO-9012) are in early prostate, renal and bladder trials.

Diagram to draw: telomere-to-cancer flowchart with two barriers.

Key points to remember:

- Telomerase = hTERT (catalytic) + hTR (RNA template).
- About 100 base pairs lost per division; 85–90% of tumours express telomerase.
- TERT promoter mutation is the commonest mutation in bladder cancer.
- TRAP assay detects urinary telomerase; inflammation gives false positives.

Q3c. Newer methods of stone analysis

Type: SAQ (10 marks) (also answered in 2024-Winter Paper 4, Q3a)

Why analyse stones

- AUA (American Urological Association) and EAU (European Association of Urology): analyse a stone at least once in every former; **repeat if stones recur on treatment** (evolving metabolic condition).
- Composition points to cause: **uric acid** suggests metabolic disease, **cystine** a genetic one, **struvite** infection. Cystine, struvite, 2,8-DHA (2,8-dihydroxyadenine) and xanthine are diagnostic in themselves.

Why wet chemical analysis is obsolete

- Detects ions, not **crystal phases**: cannot separate COM (calcium oxalate monohydrate) from COD (dihydrate), or apatite from brushite; misses rare stones.

Newer methods

Method	Principle	Comment
FTIR (Fourier-transform infrared) spectroscopy	Each compound absorbs a characteristic infrared spectrum	Commonest reference method; tiny sample; detects proteins, drugs
XRD (X-ray diffraction)	Crystal lattice diffracts X-rays in a unique pattern	Gold standard for crystals; misses amorphous material
Polarising microscopy	Crystal birefringence	Cheap; needs experience
Morpho-constitutional analysis (Daudon)	Stereomicroscopy of surface and section, plus FTIR	Subtypes point to cause, e.g. Ic suggests primary hyperoxaluria
SEM (scanning electron microscopy) ± EDX (energy-dispersive X-ray)	Surface and elements	Research
Micro-CT (micro-computed tomography), Raman spectroscopy	3-D structure; scattered light	Research

In vivo prediction

- **DECT (dual-energy computed tomography):** two energies (e.g. 80 and 140 kVp) separate **uric acid from non–uric acid** stones with near-perfect accuracy.
- **Hounsfield units:** SWL (shock wave lithotripsy) works best below about 900–1000 HU; cystine and brushite are dense.
- Dusting loses the specimen: collect fragments.

Significance in management

- **Uric acid:** oral dissolution by alkalinisation (potassium citrate).
- **Struvite:** complete clearance, culture-directed antibiotics.
- **Cystine:** hydration, alkalinisation, tiopronin; avoid SWL.
- **COM, brushite:** hard, SWL-resistant. Brushite can convert from calcium oxalate after SWL or infection.
- **2,8-DHA:** APRT (adenine phosphoribosyltransferase) deficiency; allopurinol or febuxostat. **Drug stones** (indinavir, triamterene): stop the drug.

Diagram to draw: FTIR spectrum outline; COM (dumbbell) and COD (envelope) crystals.

Key points to remember:

- FTIR and XRD are reference methods; wet chemistry is obsolete.
- DECT separates uric acid stones in vivo.
- Composition guides dissolution, SWL choice and prevention.

Q3d. Describe newer diagnostic modalities for Gu Kochs. Briefly outline role of all present diagnostic tests and give diagnostic algorithm.

Type: SAQ (10 marks) (also answered in 2023-Winter Paper 3, Q3f)

Assumed meaning: "Gu Kochs" = genitourinary tuberculosis.

Principle

- GUTB (genitourinary tuberculosis, TB) is paucibacillary with intermittent shedding: send **3–5 early-morning urine samples (EMU)** on consecutive days, processed at once.
- **Hierarchy of proof:** culture or DNA → caseating granuloma → response to treatment.

Newer modalities

- **Liquid culture, MGIT (mycobacteria growth indicator tube):** about 10 days versus 4–6 weeks on LJ (Löwenstein–Jensen).
- **NAAT (nucleic acid amplification test): GeneXpert MTB/RIF** (Xpert Ultra more sensitive): automated cartridge PCR (polymerase chain reaction), about 2 hours; detects **rifampicin resistance** (MDR (multidrug-resistant) TB surrogate). Also TrueNat; **LPA (line probe assay)** for isoniazid resistance.
- **Urine LF-LAM (lateral-flow lipoarabinomannan):** strip, under 30 minutes; WHO (World Health Organization) 2019: HIV (human immunodeficiency virus) with symptoms, advanced disease or CD4 under 200.
- **IGRA (interferon-γ release assay):** unaffected by BCG (bacille Calmette–Guérin).
- **Imaging:** CT (computed tomography) urography; **MR (magnetic resonance) urography with DWI (diffusion-weighted imaging)** grades fibrosis.

Role of present tests

Test	Role
Urine routine	**Sterile acid pyuria** ± haematuria
ZN (Ziehl–Neelsen) smear	Quick; sensitivity under 50%
Culture (gold standard)	Diagnosis, DST (drug-susceptibility testing), **monitors treatment**
NAAT	Rapid diagnosis and resistance; stays positive after cure, so cannot monitor
TST (tuberculin skin test) / IGRA	Prove infection, not active disease
KUB (kidney–ureter–bladder) film	Calcification 24–44%; "putty" kidney
IVU (intravenous urography)	Moth-eaten calyx, pipestem ureter; replaced by CT
Ultrasound	Children, pregnancy, scrotum; follows hydronephrosis
Cystoscopy and biopsy	Golf-hole orifice; biopsy if culture negative (yield 19–52%)
Chest radiograph	Mandatory (lung TB in up to one-fifth)

Diagnostic algorithm

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Suspicion (sterile pyuria, storage symptoms, epididymal nodule)
-> 3-5 EMU: ZN + culture (LJ + MGIT) + Xpert
-> chest X-ray; CT urography
Positive -> treat; DST guides regimen
Negative -> cystoscopy/biopsy, aspiration + NAAT on tissue
Still negative -> empirical trial if clinically compatible
```

Diagram to draw: algorithm above.

> **Key points to remember:**
>
> - Culture: gold standard; the only test that monitors treatment.
> - Xpert: 2 hours, rifampicin resistance; LF-LAM for HIV with CD4 under 200.
> - TST and IGRA show infection, not active disease.

Q3e. Present status of varicocele treatment

Type: SAQ (10 marks) (related: 2018-Summer Paper 3, Q3f)

Background

- Dilated pampiniform veins: about **15% of men, 35% with primary and 75–81% with secondary infertility**; mostly left (vein enters renal vein 8–10 cm higher, valve incompetence). Isolated right or non-decompressing varicocele → image for a mass.
- **Injury:** heat (lost countercurrent exchange), oxidative stress and sperm DNA fragmentation; progressive, duration-dependent; lower testosterone.
- **Grades:** subclinical (Doppler only), I (palpable with Valsalva), II (palpable), III (visible).

Current indications

Situation	Present position
Infertile couple, **palpable** varicocele, abnormal semen	**Repair** (AUA/ASRM (American Urological Association/American Society for Reproductive Medicine)). Madgar RCT (randomised controlled trial, 1995): pregnancy 44% vs 10% at 1 year
Subclinical	**Do not treat**
Before IVF/ICSI (in vitro fertilisation/intracytoplasmic sperm injection)	Improves outcomes; may allow less invasive ART (assisted reproductive technology)
NOA (non-obstructive azoospermia)	No definitive evidence
High DNA fragmentation; low testosterone	Repair improves both
Pain	Relief in most
Adolescent	Observe; repair for testicular hypotrophy over 20%, pain, or abnormal semen (Tanner 5 or 18 years)

Techniques

Technique	Recurrence	Hydrocele	Note
Microsurgical subinguinal/inguinal	**0.5–1%**	**About 0**	**Preferred**: artery, lymphatics preserved
Retroperitoneal (Palomo)	15–25%	About 7%	Children (artery ligation acceptable)
Laparoscopic	3–15%	12%	Bilateral; serious morbidity possible
Embolisation / sclerotherapy	15–25%	0	Outpatient; technical failure

- **Microsurgical steps:** micro-Doppler maps the artery; ligate all veins except vasal; preserve artery and lymphatics; deliver testis to ligate gubernacular and external spermatic veins.
- **Complications:** hydrocele (lymphatic ligation, commonest after non-microsurgical repair); artery injury (atrophy); recurrence (diagnose by upright colour Doppler).
- **Predictors of poor result:** recurrence, artery ligation, hydrocele. Earlier repair recovers more.

Diagram to draw: cord cross-section at microsurgery: artery, lymphatics and vas preserved.

Key points to remember:

- Repair only palpable varicocele with abnormal semen; never subclinical.
- Madgar 1995: pregnancy 44% vs 10%.
- Microsurgical subinguinal: recurrence under 1%, hydrocele nil.
- Adolescents: observe; repair for asymmetry over 20%.

Q3f. Recent advances in nephron sparing surgery. Role of neoadjuvant therapy.

Type: SAQ (10 marks) (NSS also answered in 2022-Summer Paper 3, Q1)

Background

- NSS (nephron-sparing surgery)/PN (partial nephrectomy): complete tumour removal, maximal parenchyma preserved. Standard for cT1a; extended to cT1b–T2 when feasible.
- **Concept:** preserved parenchymal volume, not warm ischaemia time, predicts function.

Recent advances

Area	Advance and evidence
Planning	R.E.N.A.L. (Radius, Exophytic, Nearness, Anterior/posterior, Location) and PADUA (Preoperative Aspects and Dimensions Used for an Anatomical) scores; **3D (three-dimensional) models**: randomised trial showed shorter operation and ischaemia, less blood loss
Platform	**RAPN (robot-assisted PN):** shorter warm ischaemia than laparoscopic; less blood loss than open. **Single-port robot:** shorter stay, less opioid
Vascular control	Off-clamp, selective or super-selective clamping (guided by **NIRF (near-infrared fluorescence)** with indocyanine green), early unclamping. **Randomised trials: no long-term eGFR (estimated glomerular filtration rate) gain**
Excision	**Enucleation** along the pseudocapsule: less blood loss; hereditary, multifocal, solitary kidney
Reconstruction	Sliding-clip and **barbed suture** renorrhaphy; single-layer gives better function, fewer pseudoaneurysms; sutureless with sealants in selected
Outcome reporting	**Trifecta:** negative margin + no major complication + warm ischaemia ≤25 minutes

- **Also know:** hypothermia and mannitol: no benefit with a normal contralateral kidney.

Role of neoadjuvant therapy

- **Aim:** downsize a tumour so that PN becomes possible, mainly an **imperative indication** (solitary kidney, bilateral tumours) where radical nephrectomy means dialysis.
- **TKIs (tyrosine kinase inhibitors):** pazopanib, sunitinib, axitinib. Tumour volume falls **10–30%**; resection becomes possible in **40–45%**, including PN candidates. Best in clear cell RCC (renal cell carcinoma).
- **Toxicity:** serious adverse events about 20%, mainly **wound healing**. Stop TKIs several half-lives before and after surgery.
- **Checkpoint inhibitors:** nivolumab is safe but shrinks little; trials ongoing.
- **Status:** not standard (trials only); **upfront surgery whenever feasible**. Adjuvant pembrolizumab (KEYNOTE-564) suits high-risk resected disease.

Diagram to draw: RAPN with selective segmental clamp and renorrhaphy.

> **Key points to remember:**
>
> - Parenchyma preserved, not clamp time, governs function.
> - Off-clamp and super-selective clamping: no long-term eGFR gain in randomised trials.
> - 3D models: randomised benefit; trifecta warm ischaemia ≤25 minutes.
> - Neoadjuvant TKI shrinks tumours 10–30%; for imperative PN only.

Rapid Revision Sheet – Practice Set 6 Paper 3

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

Q1. Discuss the management of B/l VUR and discuss its long term outcome (LAQ)

- Bilateral grade IV persists most; RIVUR: CAP halves UTI, not scarring.
- Cohen (over 98%) suits bilateral VUR; bilateral extravesical repair risks retention up to 10%.
- 34–38% of scarred children become hypertensive; correction does not lower BP.
- Scars bring 8× pre-eclampsia risk: the scar, not the reflux, matters.

Q2. Discuss development of urethra. Classify congenital anomalies and discuss management of epispadias (LAQ)

- Whole male urethra is endodermal UGS; fold fusion complete at 16, urethra at 18 weeks.
- About 70% of male epispadias are penopubic and incontinent; diastasis over 3 cm → exstrophy pathway.
- Cantwell–Ransley at 6–12 months; YDL at about 100 mL capacity.
- Continence after YDL: males 47–82%, females 70–87%.

Q3a. Various lasers in treatment of BPH (SAQ)

- Penetration: thulium 0.2, holmium 0.4, KTP 0.8 mm, Nd:YAG 1 cm.
- HoLEP is size-independent and superior to TURP.
- GOLIATH: PVP non-inferior to TURP; avoid PVP above 80 cc.

Q3b. Telomerase (SAQ)

- Telomerase = hTERT (catalytic) + hTR (RNA template).
- About 100 base pairs lost per division; 85–90% of tumours express telomerase.
- TERT promoter mutation is the commonest mutation in bladder cancer.
- TRAP assay detects urinary telomerase; inflammation gives false positives.

Q3c. Newer methods of stone analysis (SAQ)

- FTIR and XRD are reference methods; wet chemistry is obsolete.
- DECT separates uric acid stones in vivo.
- Composition guides dissolution, SWL choice and prevention.

Q3d. Describe newer diagnostic modalities for Gu Kochs. Briefly outline role of all present diagnostic tests and give diagnostic algorithm. (SAQ)

- Culture: gold standard; the only test that monitors treatment.
- Xpert: 2 hours, rifampicin resistance; LF-LAM for HIV with CD4 under 200.
- TST and IGRA show infection, not active disease.

Q3e. Present status of varicocele treatment (SAQ)

- Repair only palpable varicocele with abnormal semen; never subclinical.
- Madgar 1995: pregnancy 44% vs 10%.
- Microsurgical subinguinal: recurrence under 1%, hydrocele nil.
- Adolescents: observe; repair for asymmetry over 20%.

Q3f. Recent advances in nephron sparing surgery. Role of neoadjuvant therapy. (SAQ)

- Parenchyma preserved, not clamp time, governs function.
- Off-clamp and super-selective clamping: no long-term eGFR gain in randomised trials.
- 3D models: randomised benefit; trifecta warm ischaemia ≤25 minutes.
- Neoadjuvant TKI shrinks tumours 10–30%; for imperative PN only.